Problem Set 14

Integration by Parts II

October 12, 2026

Problem 2

(a)

$$\int (2x+3)e^x\,dx = \begin{bmatrix} u = 2x+3 & dv = e^x\,dx \\ du = 2dx & v = e^x \end{bmatrix}$$
$$(2x+3)e^x - 2\int e^x\,dx = \boxed{(2x+1)e^x}$$

(b)

$$\int x^3\sqrt{1+x^2}\,dx = \begin{bmatrix} u = x^2 & dv = x\sqrt{1+x^2}\,dx \\ du = 2x\,dx & v = \frac{\sqrt{1+x^2}^3}{3} \end{bmatrix}$$
$$\frac{x^2(1+x^2)^{\frac{3}{2}}}{3} - \int \frac{2x(1+x^2)^{\frac{3}{2}}}{3}\,dx$$
$$= \frac{x^2(1+x^2)^{\frac{3}{2}}}{3} - \frac{2}{15}(1+x^2)^{\frac{5}{2}}$$
$$= \boxed{\frac{1}{15}(1+x^2)^{\frac{3}{2}}(3x^2-2)+C}$$

Problem 4

(a)

$$\int \sin^2 x\,dx = -\frac{1}{2}\cos x\sin x + \frac{1}{2}\int 1\,dx = \boxed{\frac{x}{2} - \frac{\sin 2x}{4} + C}$$

(b)

$$\int \sin^4 x\,dx = -\frac{1}{4}\cos x\sin^3 x + \frac{3}{4}\int \sin^2 x\,dx$$
$$= -\frac{1}{4}\cos x\sin^3 x + \frac{3}{4}\left(\frac{x}{2} - \frac{\sin 2x}{4}\right)$$
$$= \boxed{\frac{3}{8}x - \frac{3}{16}\sin 2x - \frac{1}{4}\cos x\sin^3 x + C}$$

Problem 5

(a)

$$W_n = \int_0^{\frac{\pi}{2}} \sin^n x\,dx = -\frac{1}{n}\cos x \sin^{n-1} x\Big|_0^{\frac{\pi}{2}} + \frac{n-1}{n}\int_0^{\frac{\pi}{2}} \sin^{n-2} x\,dx = \frac{n-1}{n}W_{n-2}$$

(b)

$$W_0 = \int_0^{\frac{\pi}{2}} 1\,dx = \frac{\pi}{2}$$
$$W_1 = \int_0^{\frac{\pi}{2}} \sin x\,dx = 1$$
$$W_3 = \frac{2}{3}\cdot 1 = \frac{2}{3}$$
$$W_5 = \frac{4}{5}\cdot\frac{2}{3} = \frac{8}{15}$$

(c)

$$W_{2n+1} = \frac{2n}{2n+1}W_{2n-1} = \cdots = \frac{2\cdot 4\cdot 6\ldots(2n)}{3\cdot 5\cdot 7\ldots(2n+1)}$$

(d)

$$W_{2n} = \frac{2n-1}{2n}W_{2n-2} = \cdots = \frac{3\cdot 5\cdot 7\ldots(2n-1)}{2\cdot 4\cdot 6\ldots(2n)}\cdot\frac{\pi}{2}$$

The π originates in the term W_0.

(e) Fractions with values less than 1 are being multiplied.

Problem 6

(a)

$$\int \ln^n x\,dx = \begin{bmatrix} u = \ln^n x & dv = dx \\ du = \frac{n}{x}\ln^{n-1} x\,dx & v = x \end{bmatrix}$$
$$\boxed{x\ln^n x - n\int \ln^{n-1} x\,dx}$$

(b)

$$\int \ln x = x\ln x - x$$
$$\int \ln^2 x = x\ln^2 x - 2x\ln x + 2x$$
$$\int \ln^3 x = \boxed{x^3\ln x - 3x\ln^2 x + 6x\ln x - 6x + C}$$

(c)

$$\int \tan^n x\,dx = \int \tan^{n-2}(\sec^2 x - 1)\,dx$$
$$\int \tan^{n-2}\sec^2 x\,dx = \begin{bmatrix} u = \tan x \\ du = \sec^2 x\,dx \end{bmatrix} \int u^{n-2}\,du = \frac{\tan^{n-1} x}{n-1}$$
$$\int \tan^n x\,dx = \boxed{\frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x\,dx}$$

(d)

$$\int \tan x\,dx = \ln|\sec x|$$
$$\int \tan^3 x = \frac{\tan^2 x}{2} - \ln|\sec x|$$
$$\int \tan^5 x = \boxed{\frac{1}{4}\tan^4 x - \frac{1}{2}\tan^2 x + \ln|\sec x| + C}$$

Problem 7

(a)

$$s(t) = \int t^2 e^{-t}\,dt = \begin{bmatrix} u = t^2 & dv = e^{-t}\,dt \\ du = 2t\,dt & v = -e^{-t} \end{bmatrix} - t^2e^{-t} + \int 2te^{-t}\,dt$$
$$= \begin{bmatrix} u = t & dv = e^{-t}\,dt \\ du = dt & -e^{-t} \end{bmatrix} - t^2e^{-t} - 2te^{-t} + \int 2e^{-t}\,dt = -(t^2+2t+2)e^{-t}$$
$$s(t) - s(0) = \boxed{2 - (t^2+2t+2)e^{-t}}$$

(b)

$$\lim_{t\to\infty} 2 - (t^2+2t+2)e^{-t} = \boxed{2}$$

(c)

$$v(t) = t^2e^{-t}$$
$$v'(t) = 2te^{-t} - t^2e^{-t}$$
$$v'(t) = 0 \implies \boxed{t = 2}$$

Problem 9

(a)

$$f(x) - f(a) = \int_a^x f'(t)\,dt$$

$$\int_a^x f'(t)\,dt = \begin{bmatrix} u = f'(t) & dv = dt \\ du = f''(t)\,dt & v = t - x \end{bmatrix} (t-x)f'(t)\Big|_a^x - \int_a^x (t-x)f''(t)\,dt$$

$$f(x) - f(a) = (t-x)f'(t)\Big|_a^x - \int_a^x (t-x)f''(t)\,dt$$

$$f(x) = f(a) + (x-a)f'(a) + \int_a^x (x-t)f''(t)\,dt$$

$$\boxed{f(x) = L(x) + \int_a^x (x-t)f''(t)\,dt}$$

(b)

$$|f(x) - L(x)| = |\int_a^x (x-t)f''(t)\,dt|$$

$$|f''(t)| \le M \implies \int_a^x (x-t)f''(t)\,dt \le M\int_a^x x - t\,dt$$

$$M\int_a^x x - t\,dt = -\frac{M(x-t)^2}{2}\Big|_a^x = \frac{M}{2}(x-a)^2$$

$$\boxed{|f(x) - L(x)| \le \frac{M}{2}(x-a)^2}$$

(c)

$$f(x) - L(x) = \int_a^x (x-t)f''(t)\,dt$$

$$f''(t) \ge 0 \implies \int_a^x (x-t)f''(t)\,dt \ge 0$$

$$\boxed{f(x) \ge L(x)}$$

(d)

$$\int_a^x (x-t)f''(t)\,dt = \begin{bmatrix} u = f''(t) & dv = (x-t)\,dt \\ du = f'''(t)\,dt & v = -\frac{(x-t)^2}{2} \end{bmatrix}$$

$$-\frac{1}{2}(x-t)^2 f''(t)\Big|_a^x + \int_a^x \frac{1}{2}(x-t)^2 f'''(t)\,dt$$

$$\boxed{f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \int_a^x \frac{f'''(t)}{2}(x-t)^2\,dt}$$

Problem 10

(a)

$$B(m,n) = \int_0^1 x^m(1-x)^n\,dx$$

$$= \begin{bmatrix} u = (1-x)^n & dv = x^m\,dx \\ du = -n(1-x)^{n-1} & v = \frac{x^{m+1}}{m+1} \end{bmatrix} \frac{1}{m+1}(1-x)^n x^{m+1}\Big|_0^1$$
$$+ \int_0^1 \frac{n}{m+1} x^{m+1}(1-x)^{n-1}$$
$$= \boxed{\frac{n}{m+1} B(m+1, n-1)}$$

(b)

$$\begin{aligned} B(m,n) &= \frac{n}{m+1} B(m+1, n-1) \\ &= \frac{n}{m+1} \cdot \frac{n-1}{m+2} B(m+2, n-2) \\ &= \frac{n}{m+1} \cdot \frac{n-1}{m+2} \cdot \frac{n-2}{m+3} B(m+3, n-3) \\ &\cdots \\ &= \frac{n(n-1)\cdots 1}{(m+1)(m+2)\cdots(m+n)} B(m+n, 0) \\ &= \frac{n!}{(m+1)(m+2)\cdots(m+n)(m+n+1)} \\ &= \boxed{\frac{m!\,n!}{(m+n+1)!}} \end{aligned}$$

(c)

$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx = \int_0^1 \left(x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2} \right) dx$$
$$= \frac{x^7}{7} - \frac{2x^6}{3} + x^5 - \frac{4x^3}{3} + 4x - 4\arctan(x)\Big|_0^1 = \boxed{\frac{22}{7} - \pi}$$
$$\int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0 \implies \boxed{\pi < \frac{22}{7}}$$

(d)

$$B(4,4) = \int_0^1 x^4(1-x)^4\,dx = \frac{4!\cdot 4!}{9!} = \frac{1}{630}$$
$$\frac{1}{2} \le \frac{1}{1+x^2} \le 1$$
$$\frac{1}{1260} \le \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx \le \frac{1}{630}$$
$$\frac{1}{1260} \le \frac{22}{7} - \pi \le \frac{1}{630}$$
$$\frac{1979}{630} \le \pi \le \frac{3959}{1260}$$
$$\boxed{3.1413 \le \pi \le 3.1421}$$